```
    token.attributes.put("client_id",     #A
            response.optString("client_id"));    #A

    return Optional.of(token);
}
```

#A Extract token attributes from the relevant fields in the response.

Although you used the `sub` field to extract an ID for the user, this may not always be appropriate. The authenticated subject of a token needs to match the entries in the `users` and `permissions` tables in the database that define the access control lists for Natter social spaces. If these don't match, then the requests from a client will be denied even if they have a valid access token. You should check the documentation for your AS to see which field to use to match your existing user IDs.

You can now switch the Natter API to use OAuth2 access tokens by changing the `TokenStore` in Main.java to use the `OAuth2TokenStore`, passing in the URI of your AS's token introspection endpoint and the client ID and secret that you registered for the Natter API (see appendix A for instructions):

```
var introspectionEndpoint =

URI.create("https://as.example.com:8443/oauth2/introspect");
SecureTokenStore tokenStore = new OAuth2TokenStore(     #A
     introspectionEndpoint, clientId, clientSecret);    #A
var tokenController = new TokenController(tokenStore);
```